

Non-parametric estimation of the risk-neutral distribution for the SPX index from market data

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What my project is about

I calculate the 28-day risk-neutral distribution (RND) for the SPX (S&P 500) from its options (implied volatility) data downloaded from Yahoo Finance

I compare the RND to the historical data for the past 1 year, to determine how different the predicted and past distributions are for the 28-day returns

The computation and analyses are carried out in a Jupyter notebook (JulianOng_ClassProject.ipynb) which was run on November 3rd, 2025

Two crucial tools: (1) the Savitzky-Golay filter to clean the volatility data, and (2) the Breeden-Litzenberger identity to determine the risk-neutral distribution



Risk-neutral distribution

Market price for a call option with strike price K at time T is determined by the risk-neutral distribution $g(S_T)$

$$C(K) = e^{-rt} \int_K^{\infty} (S_T - K) g(S_T) dS_T$$

The RND reflects the current market (empirical) view of the future distribution of asset prices at time T

We can compute the RND using the **Breeden-Litzenberger** identity [1] and finite differences

$$g(S_T) = e^{rt} \left. \frac{\partial^2 C}{\partial K^2} \right|_{K=S_T} \approx e^{rt} \left(\frac{C(K + \delta) + C(K - \delta) - 2C(K)}{2\delta^2} \right)$$

[1] Breeden, D. T., & Litzenberger, R. H. (1978). Prices of state-contingent claims implicit in option prices. *Journal of business*, 621-651.

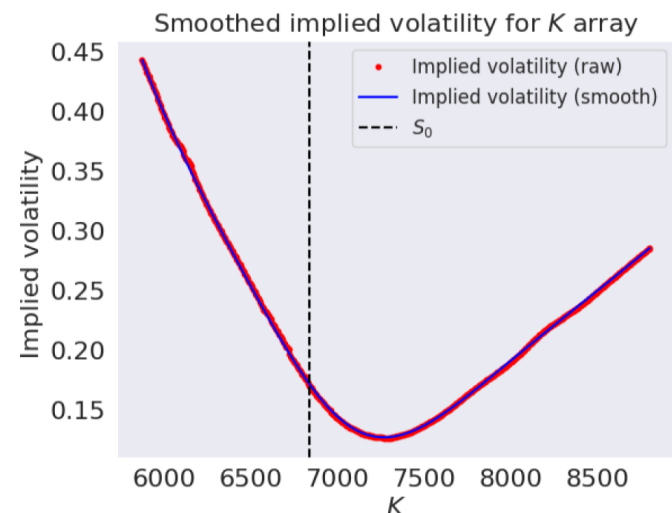
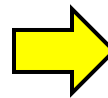
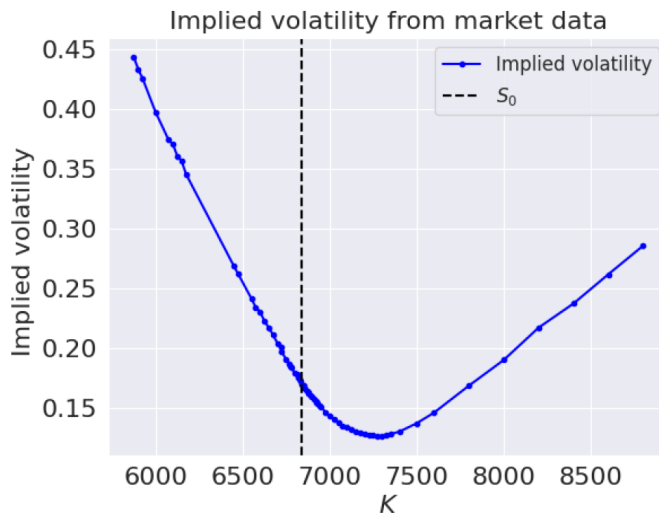


Cleaning of implied volatility data

I construct the volatility smile from the downloaded options data

I generate the volatility smile over a uniform array of strike price (K) points with spacing δ , using piecewise polynomial interpolation

I filter out the noise in the volatility smile, using **Savitzky-Golay** smoothing [1], a signal processing technique commonly used in physics and chemistry for cleaning spectroscopic data



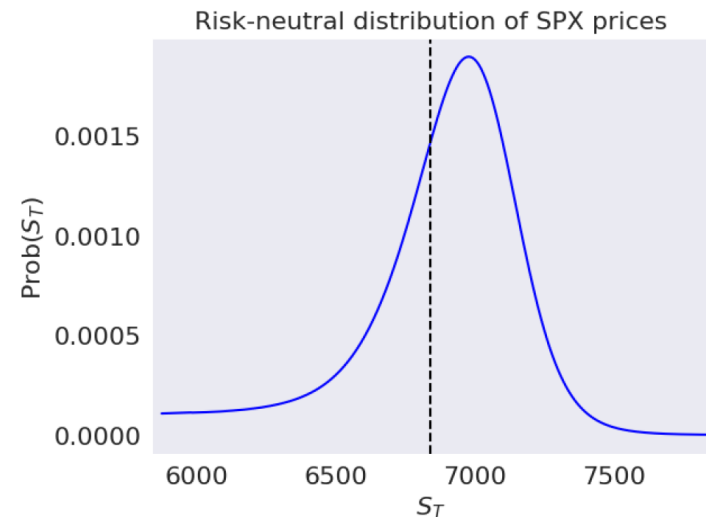
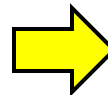
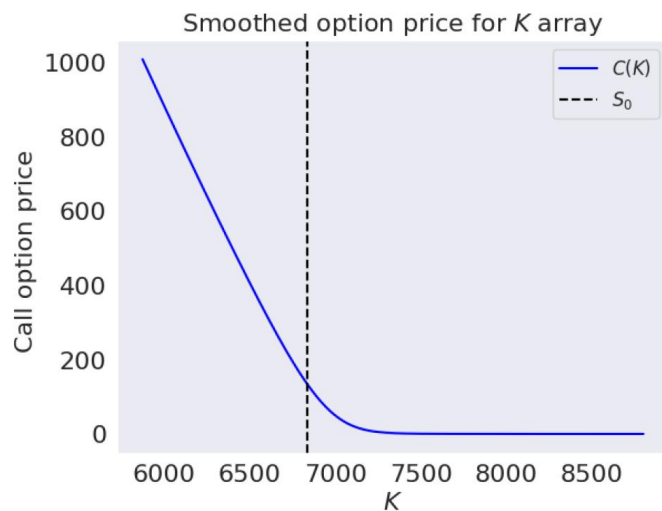
[1] Savitzky, A., & Golay, M. J. (1964). Smoothing and differentiation of data by simplified least squares procedures. *Analytical chemistry*, 36(8), 1627-1639.



Generation of the RND

I use the classic Black-Scholes formula to generate the call prices $C(K)$ over the K array from the smoothed volatility data

I generate the RND from the numerical 2nd derivative of the $C(K)$ array, using the Breeden-Litzenberger identity



The RND appears not to have a lognormal form which usually has a positive skew

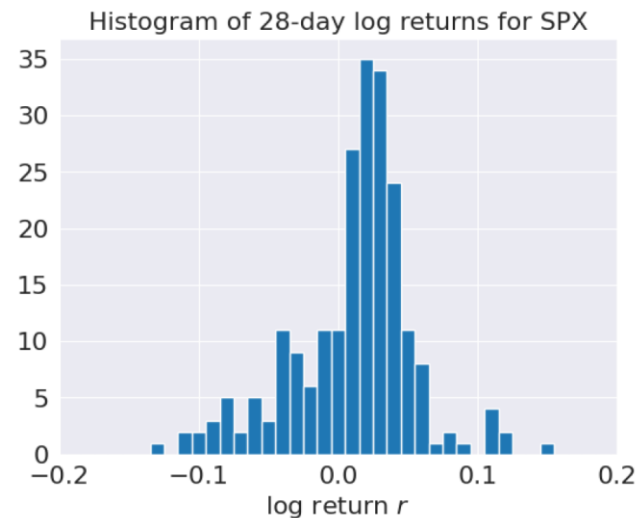
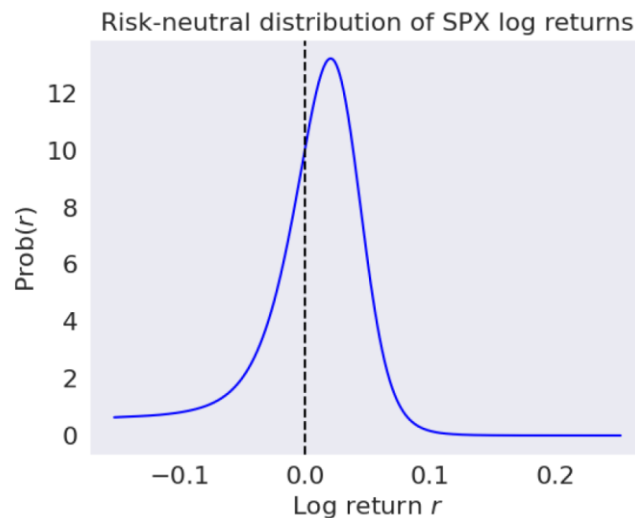


Historical log-return distribution for SPX

I compute the RND for log returns, given by $p(r) = g(S) \cdot S$

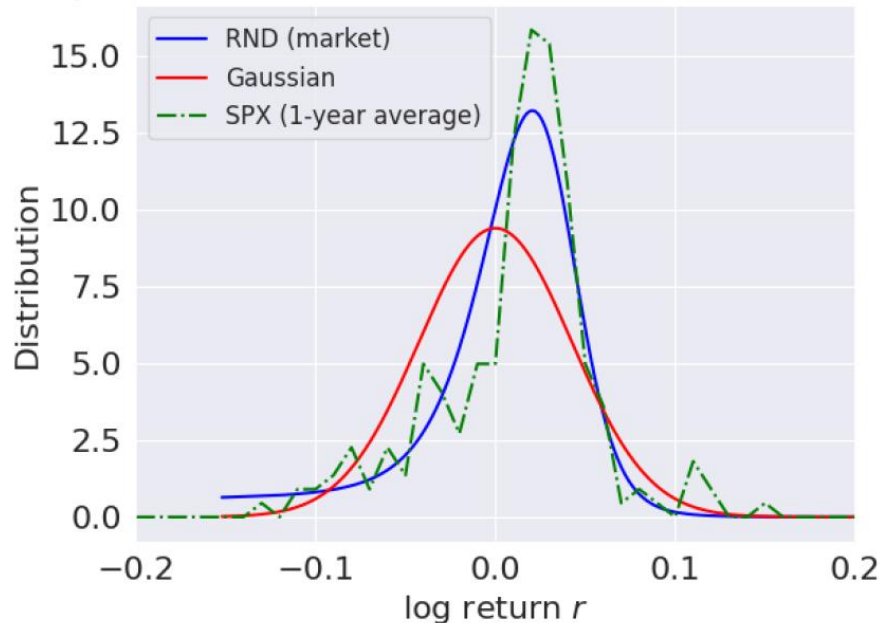
I analyze the historical distribution of the 28-day log return ($r_t = \log S_{t+T} - \log S_t$) in the past 1 year for the SPX

Both distributions exhibit negative skewness



Comparison of RND and market data

Comparison of RND with normal distribution and market data



Both the RND and the historical data for log returns are non-Gaussian and exhibit similar negative skewness (also seen in the Heston model)

This suggests that market predictions are in line with historical data for the SPX



Future work

Comparison of the RND with predictions of a parametric model such as the Heston model

Calibration of the Heston model, by fitting its characteristic function with the Fourier transform of the RND

Thank you

